

```
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
```

```
#define MAXPAROLA 30
#define MAXRIGA 80
```

```
int main(int argc, char *argv[])
{
    int freq[MAXPAROLA]; /* vettore di contatori
delle frequenze delle lunghezze delle parole */
    char riga[MAXRIGA];
    int i, inizio, lunghezza;
    FILE *f;
```

```
for(i=0; i<MAXPAROLA; i++)
    freq[i]=0;
```

```
if(argc != 2)
```

```
{
    printf(stderr, "ERRORE, serve un parametro con il nome del file\n");
    exit(1);
}
```

```
f = fopen(argv[1], "r");
if(f==NULL)
```

```
{
    printf(stderr, "ERRORE, impossibile aprire il file %s\n", argv[1]);
    exit(1);
}
```

```
while( fgets( riga, MAXRIGA, f ) != NULL )
```



System and Device Programming

Introduction to SDP

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Politecnico di Torino

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General Information

❖ System and Device Programming

➤ 01NYH_{OV}, ING-INF/05

❖ Master of science degree

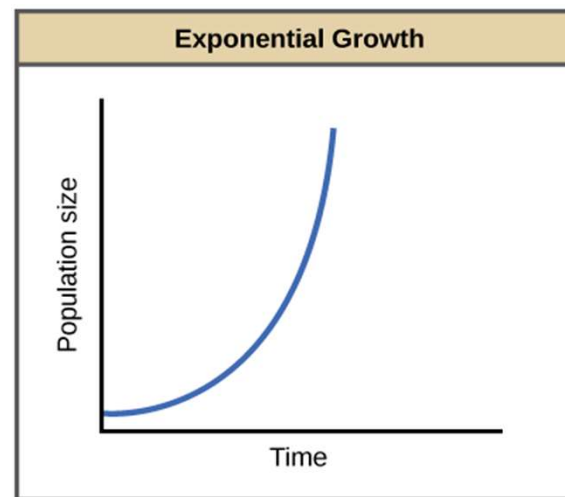
➤ Computer Engineering

- 1^o year, 10 credits, **100** hours
- All students (from A to Z)

SDP 5th Edition

Similar to edition 4

Number of students enrolled
A.Y. 2019-2020 – 137 (265)
A.Y. 2020-2021 – 134 (310)
A.Y. 2021-2022 – 140 (318)
A.Y. 2022-2023 – 160 (320)
A.Y. 2023-2024 – 169 (376)



General Information

- ❖ System and Device Programming
 - 01NYH_{OV}, ING-INF/05
- ❖ Master of science degree
 - Computer Engineering
 - 1^o year, 10 credits, **100** hours
 - All students (from A to Z)
- ❖ The course includes 2 separate parts
 - Operating system design and internal features
 - System and device programming

SDP 5nd Edition

Similar to edition 4

First part

Second part

Instructors

First part

❖ Operating System Design

➤ Gianpiero Cabodi

- Dept. of Control and Computer Engineering
- Phone: 011 090 - 7082
- E-mail: gianpiero.cabodi@polito.it



**Politecnico
di Torino**

Department of Control and
Computer Engineering



Instructors

Second part

❖ System and Device Programming

➤ Stefano Quer

- Dept. of Control and Computer Engineering
- Phone: 011 090 - 7076
- E-mail: stefano.quer@polito.it



➤ Michelangelo Grosso

- Dept. of Control and Computer Engineering
- E-mail: michelangelo.grosso@polito.it



**Politecnico
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Department of Control and
Computer Engineering



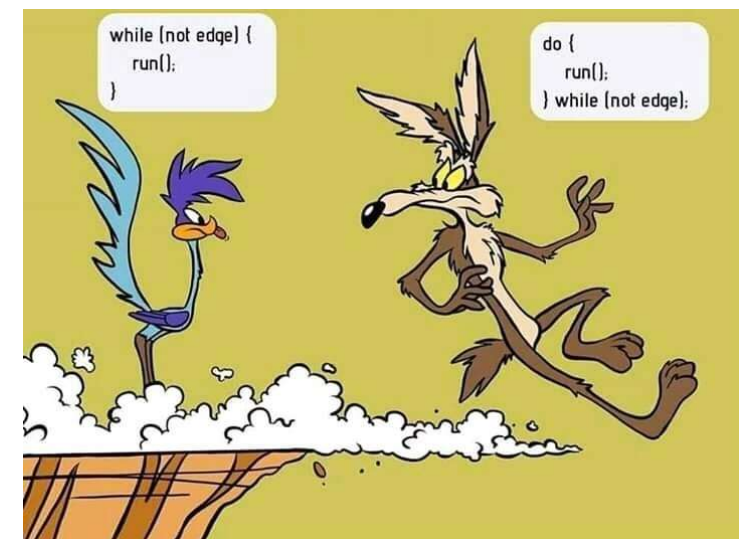
Background Requirements

- ❖ The course is **incremental** with respect to
 - All programming-oriented classes at the BS level
 - The class of Operating Systems at the BS level
- ❖ **Strict** prerequisites in terms of
 - **Programming skills**
 - C language syntax
 - Problem solving including dynamic memory allocation, recursion, abstract data types (lists, trees, heaps, graphs)
 - IDEs under the Unix OS
 - For example, gcc, g++, gdb, codeBlock, CLion, etc.

Background Requirements

➤ **Operating system** basic features

- UNIX/Linux Operating System
- Processes
- Threads
- Synchronization paradigm and related primitives
- Elementary parallel programming with UNIX processes and POSIX threads



Learning Outcomes

- ❖ Acquire **adequate** knowledge of
 - Operating system internal features
 - System, device, and parallel programming
- ❖ Develop applications
 - To mimic the main features of real operating systems
 - Under different
 - Environments (UNIX-like systems, Windows API, MacOS)
 - Standards (C, POSIX, C++)

First part
(50%)

Second part
(50%)

Delivery Mode

- ❖ There is no distinction between theory lectures and practice lessons
 - Lectures include practice lessons
 - 3 (+1 extra) blocks of 1.5 hours every week
 - Monday, 11.30-13.00, classroom **R2**
 - Tuesday, 13.00-14.30, classroom **R2**
 - Wednesday, 11.30-14.30, classroom **P1**
 - One is a recovery slot
 - Used only during specific weeks (when necessary)

Delivery Mode

❖ Lectures and practice are extended with (about) 30 additional hours in laboratory

- 4 blocks
- 2 blocks of 1.5 hours for each lab team
- Team A: A - F , Monday 14.30-16.00, labINF
- Team B: G - Z , Friday 16.00-19.00, labINF

Not done during weeks
1 and 2

Delivery Mode

- ❖ The course is video-recorded and opens a virtual classroom
- ❖ It is divided into two parts
 - **First part**
 - Theoretical aspects of OS design
 - Laboratories
 - Analysis and implementation of operating system internal features

More details on the first part will be given by Prof. Cabodi

Delivery Mode

Include **parallel programming, system programming, problem solving**

➤ Second part

- System and device programming
 - Cross-platform programming in C++
 - UNIX/Linux (C & POSIX) programming
- Laboratories
 - Problem solving
 - From specs to code through editing, compilation, debugging, and execution of programs
 - Fix defects, fill small missing code sections, improve programs, solve classical problems
 - Every week
 - You will find a new laboratory test
 - One or more questions/exercises
 - You will get the solutions of the previous lab

Delivery Mode

➤ Second part

Include **parallel programming, system programming, problem solving**

- System and device programming
 - Cross-platform programming in C++
 - UNIX/Linux (C & POSIX) programming
- Laboratories

Labs are mandatory
(as lectures)

They are part of the course and
there may be specific questions
during the exams targeting
laboratory topics



Each of you will need

A Linux machine with all main apps

Tentative Plan

- ❖ Check the "Portal Web Page" for
 - A more detailed version of the schedule
 - Changes and updates

3 + 1 extra slots
2 lab slots

First part
(50%)

Second part
(50%)

More details will
follow

Week	Starting date	Ending date	Instructors	Comments
1	4 March	10 March	Quer	No laboratory
2	11 March	17 March	Cabodi	No laboratory
3	18 March	24 March	Quer	
4	25 March	31 March	Cabodi	
5	08 April	14 April	Quer	
6	15 April	21 April	Cabodi	
7	22 April	28 April	Quer	
8	29 April	05 May	Cabodi	
9	06 May	12 May	Quer	
10	13 May	19 May	Cabodi	
11	20 May	26 May	Quer	
12	27 May	02 June	Cabodi	
13	03 June	07 June	Quer	
14	10 June	16 June	Cabodi	

Tentative Plan

- ❖ Check the “Portal Web Page” for
 - A more detailed version of the schedule
 - Changes and updates

3 + 1 extra slots
2 lab slots

First part
(50%)

Week	Starting date	Ending date	Instructors	Comments
1	4 March	10 March	Quer	Review
3	18 March	24 March	Quer	C++ (basics)
5	08 April	14 April	Quer	C++ (libraries)
7	22 April	28 April	Quer	Synch (POSIX)
9	06 May	12 May	Quer	Synch (C++)
11	20 May	26 May	Quer	IO
13	03 June	07 June	Quer	IPC

Texts, Readings, Handouts

❖ General information (teaching portal)

- Calendar, rules and deadlines
- Exams bookings and exam results

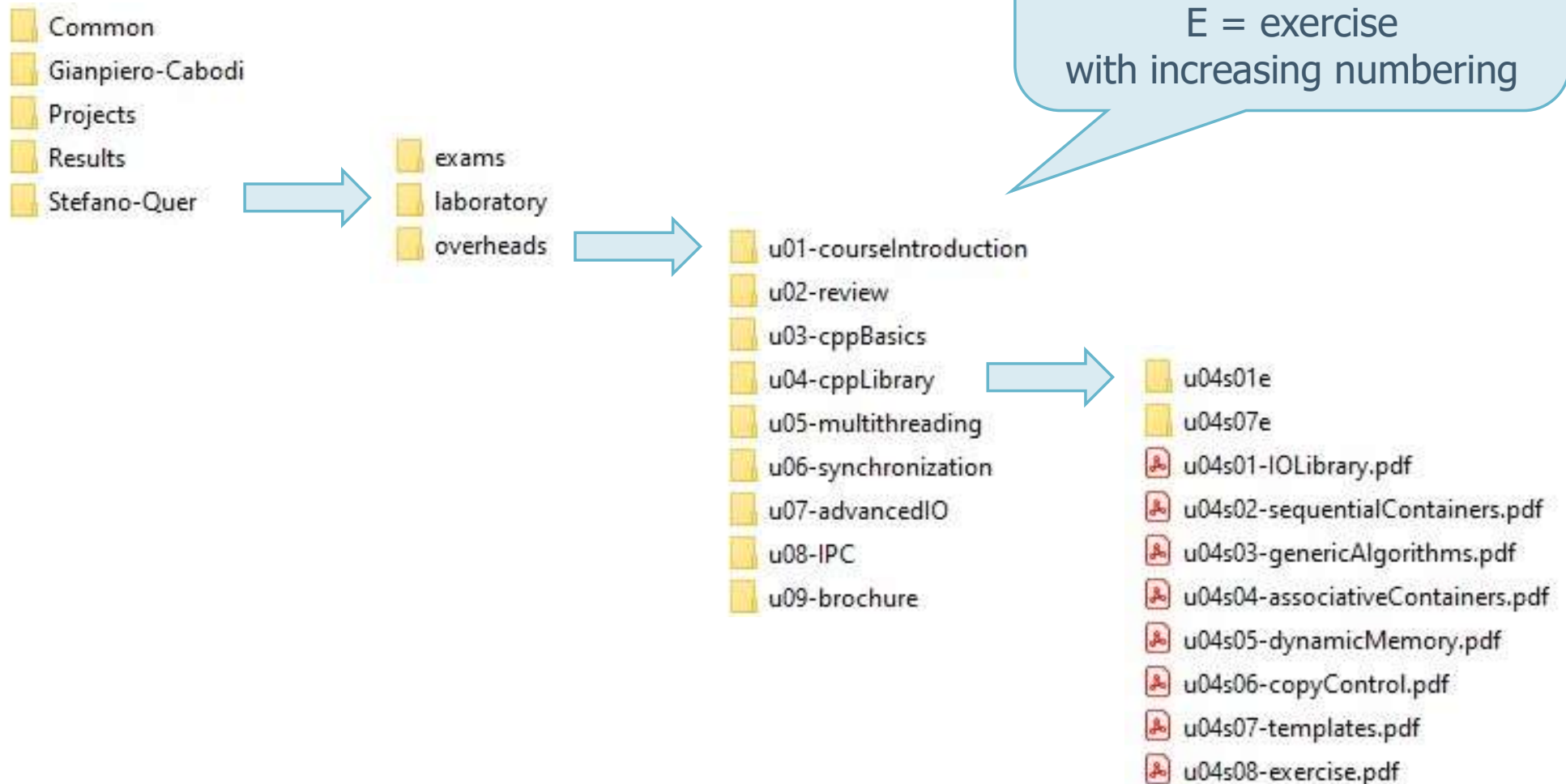
❖ Material used

- On **Dropbox** on the Politecnico portal page
- Overheads
- Laboratory
 - 1 specification every week
 - 1 solution every week (previous week)
- Examination texts
 - All "compatible" versions are made available
 - Solution (almost all theory solved)

Possibly, a Slack channel to reply to common questions/problems

Delivery Mode

Slides are numbered as
U = unit
S = section
E = exercise
with increasing numbering



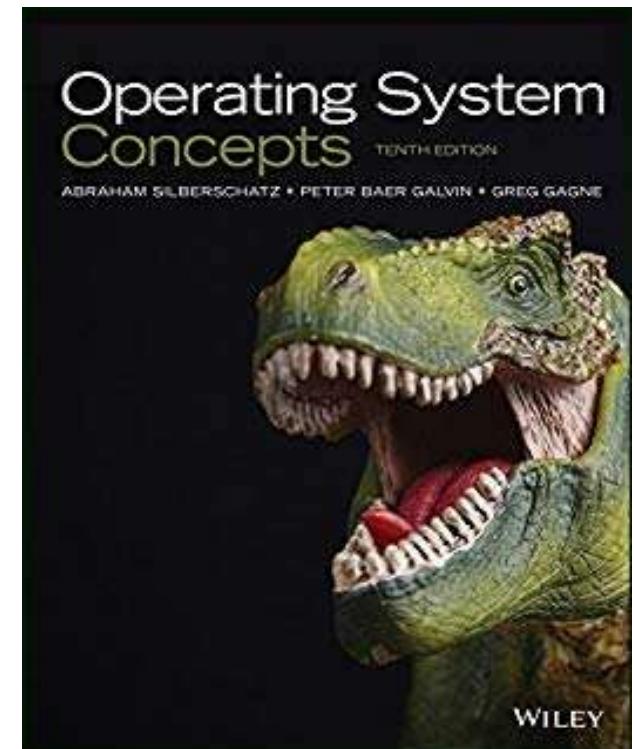
Texts, Readings, Handouts

❖ Printed material

➤ Theory part

- A. Silberschatz, P. B. Galvin, G. Gagne, "Operating System Concepts", 10th edition, Wiley, 2019, ISBN: 978-1119456339

Reference book for all
theoretic aspects on
operating systems



Texts, Readings, Handouts

❖ Printed material

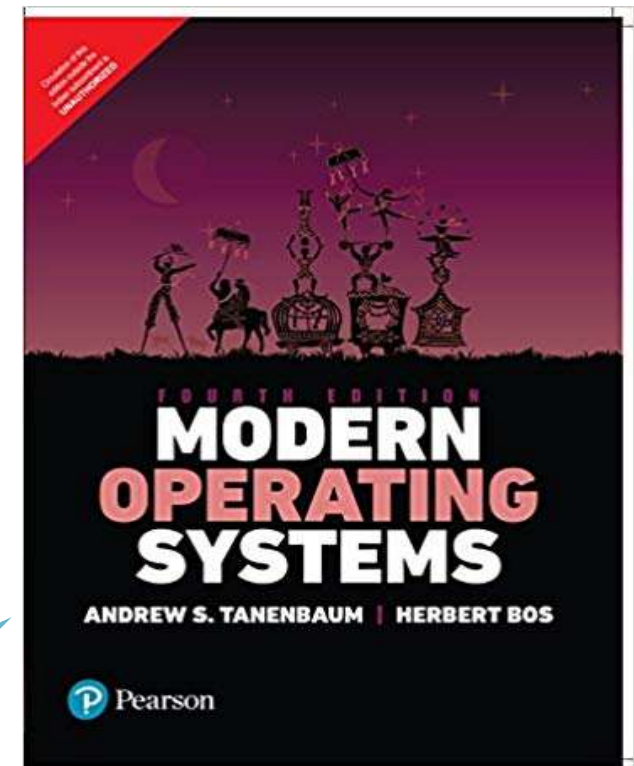
➤ Theory part

- Andrew S. Tanenbaum, "Modern Operating Systems", 4th edition, Pearson, 2019, ISBN: 978-9332575776

Good alternative to the previous reference



5° edition



4° edition

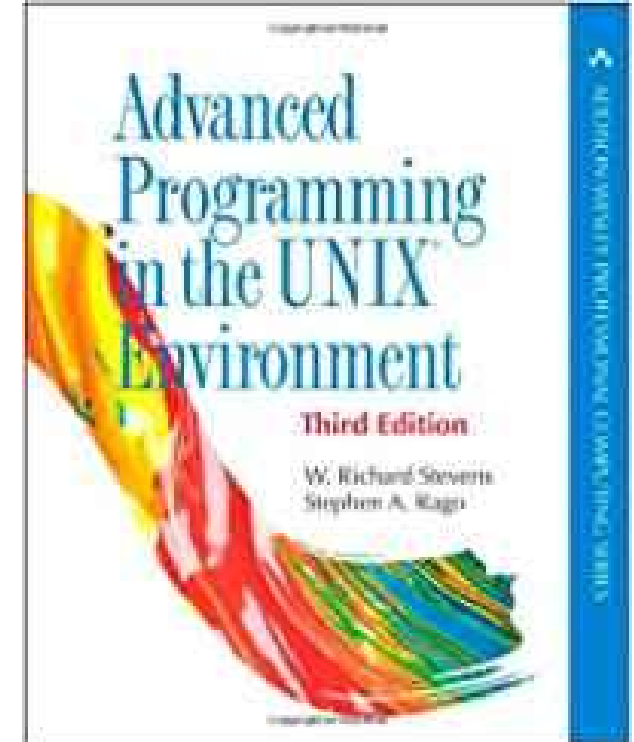
Texts, Readings, Handouts

❖ Printed material

➤ Programming part

- W. R. Stevens, S. A. Rago, "Advanced programming in the UNIX Environment", 3rd edition, Addison-Wesley, 2013, ISBN: 978-0-321-63773-4

Reference book for all aspects on UNIX-like systems



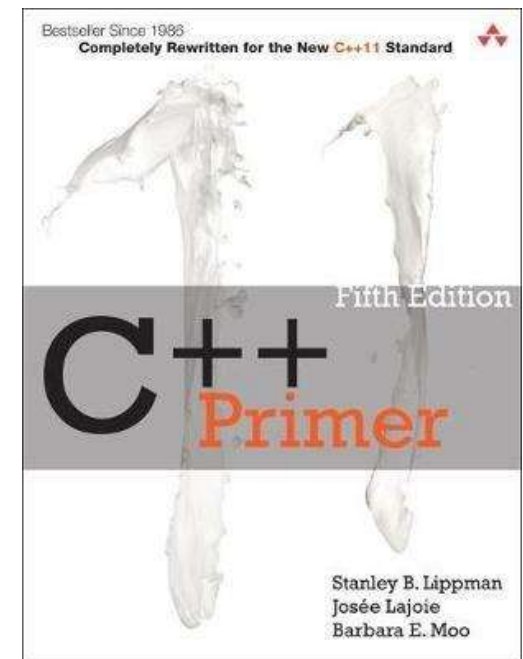
Texts, Readings, Handouts

❖ Printed material

➤ Programming part

- Stanley B. Lippman, Josée Lajoie, Barbara E. Moo, "C++ Primer", Addison Wesley Professional, ISBN 978-0-321-71411-4

Possible reference book
for C++ programming



Texts, Readings, Handouts

❖ Other material

- Additional information, clarifications, and more are available on the WWW
 - Implementation details
 - Libraries
 - Specs
 - Data types, system calls
 - etc.

Assessment and Grading Criteria

❖ **Premises** for students who followed the course before 2021-2022

- Since June 2021, the old examination format (following the Prof. Laface's program) has **not** been available anymore
- All candidates **must** follow the **new** program and the **new** examination rules
 - Old students can take the project **if and only if** they are moving to the new exam format for the first time, i.e., they did never take the exam with the new format during previous years
 - **Warning:** Projects are knowledge-demanding and time-consuming; go for them with great care

Assessment and Grading Criteria

❖ The exam consists in

➤ A written test

- To check the
 - Knowledge on theory topics
 - Problem-solving ability

➤ An **optional** group project

- To improve and verify the
 - Capacity to face large and complex realistic problems and tasks
 - Ability to work in groups with colleagues

On-Off



Standard



Assessment and Grading Criteria

- ❖ The written test is divided into two sessions
 - The first one is on operating system internals
 - The second one is on system and device programming

First part
(50%)



Second part
(50%)



- ❖ Each session

- Last from 60 to 120 minutes, depending on the examination text
- Must be completed **with no** books, notes, or other printed or electronic material
- Portable devices (e.g., cell phones) must be turned-off during the entire test



Standard

Assessment and Grading Criteria

- ❖ Both sessions include questions and exercises on both theoretical and practical aspects of the relative part of the course
 - Theoretical questions may include open and closed puzzles
 - These are formulated to test the candidate's knowledge on all topics presented
 - Practical questions may include programming exercises
 - These test the ability of the candidate to write programs in different environments and programming languages (UNIX/Linux, C and C++)

Assessment and Grading Criteria

❖ Evaluation

- Each part amounts **up to** 15 points on the final mark and it has a **passing threshold** of 7 points
- The final mark is the sum of the two evaluations
- To pass the exam (with or without the group project)
 - A pass grade is required on both parts
 - Each mark must be larger or equal than 7
 - A pass grade is required on their sum
 - The sum must be larger or equal than 18
- The maximum grade reachable with the written test is equal to $15+15=30/30$

Assessment and Grading Criteria

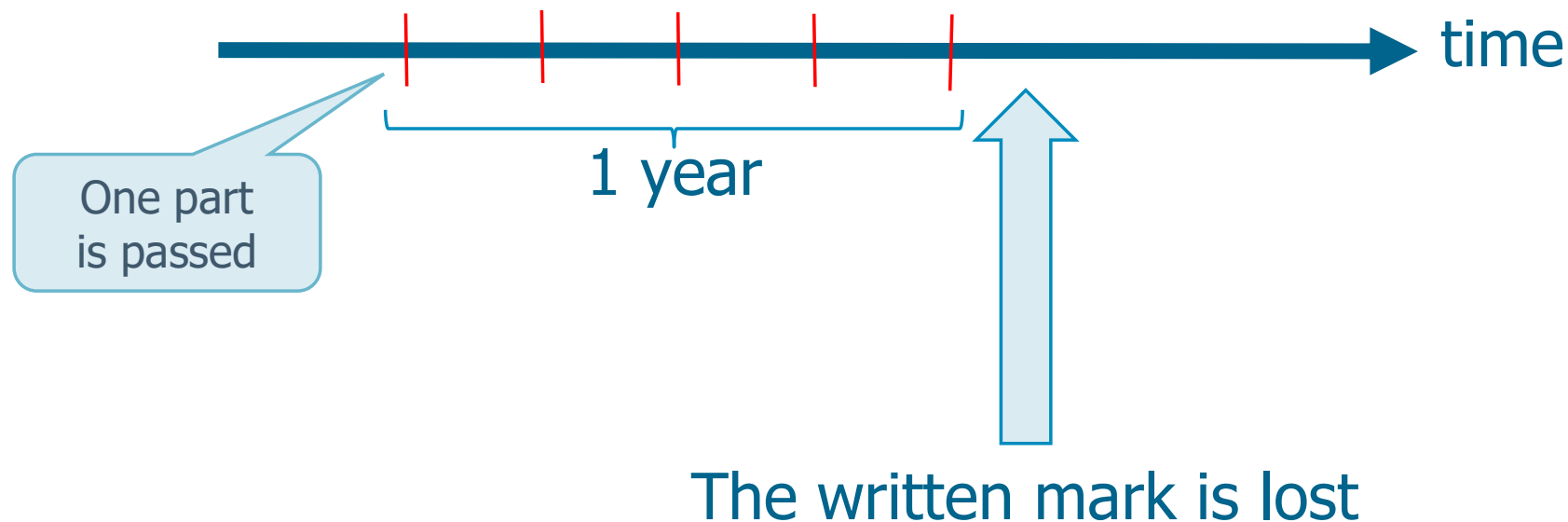
❖ The two written parts

- Can be taken during different examination sessions
 - Marks can be rejected to reach the passing threshold or improve the final sum/mark
- These sessions must belong to a time period corresponding to at most 4 (+1) examination sessions

Extra (autumn)
session

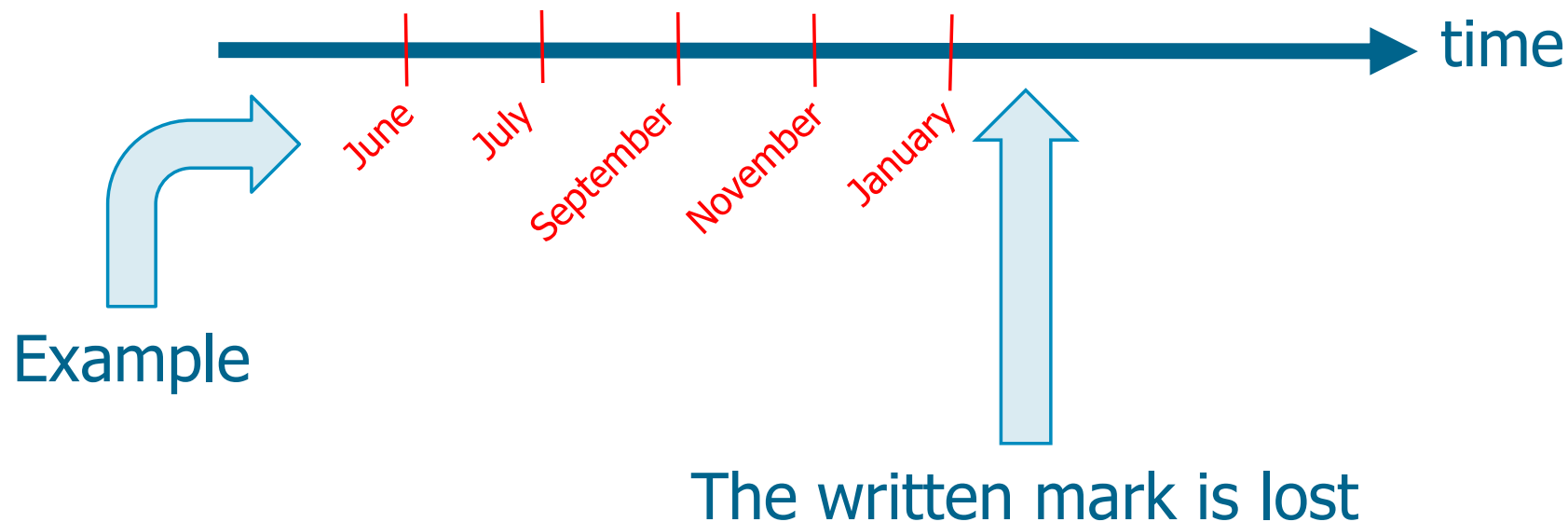
Assessment and Grading Criteria

- When the **time window** is expired the written part that has been passed but not finalized (registered with a final mark) will be automatically lost and the candidate will have to retake the corresponding written test



Assessment and Grading Criteria

- When the **time window** is expired the written part that has been passed but not finalized (registered with a final mark) will be automatically lost and the candidate will have to retake the corresponding written test



Assessment and Grading Criteria

- ❖ An **optional** project may be taken to
 - Improve the candidate's knowledge on the course's topics
 - Eventually, amend the final mark
- ❖ In any case
 - The exam is passed **only** if the written test is passed
 - A project can be taken only **once** by each students
 - The project must be selected **before** taking any written examination tests

Projects are **not** made to improve the final mark
Projects are made **to learn more**

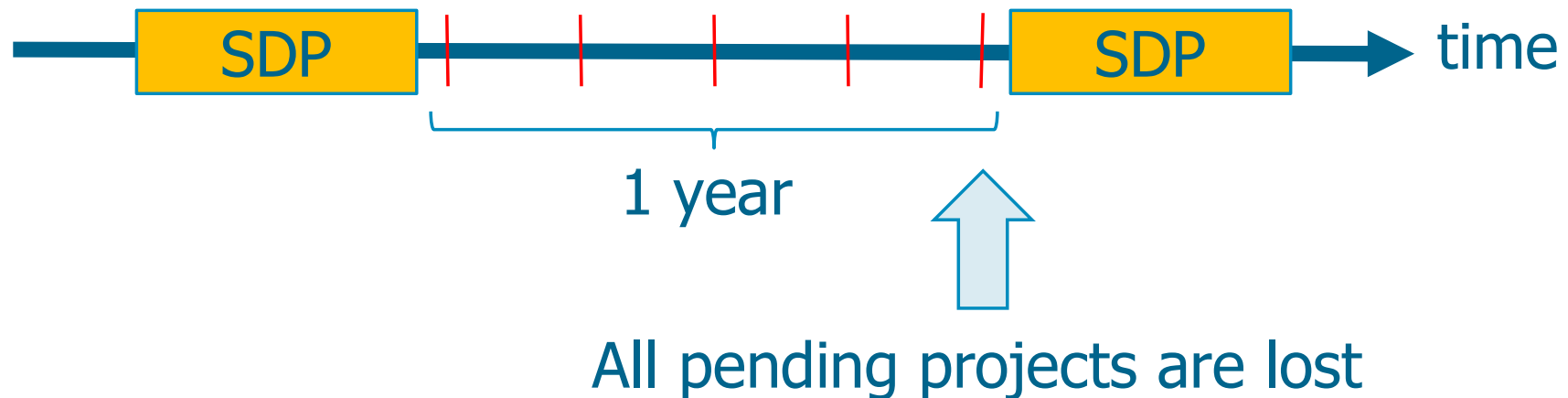
Assessment and Grading Criteria

❖ Projects

- Are specified as soon as possible during the course but not before students can appreciate/understand them
- May have as a subject the “operating system internals” or “system and device programming”
- Must be selected by group of 2 or 3 candidates
- Will be assigned to group of students on a **first-come-first-served** basis

Assessment and Grading Criteria

- Must be completed and delivered within the same academic year (i.e., in one of the 4+1 examination sessions following each course)
 - The beginning of a new course will automatically cancel all pending projects



Assessment and Grading Criteria

❖ Project evaluation

- The evaluation will follow a short presentation (15-20 minutes) given by the group of candidates during an examination session following the written tests
- To enroll for the presentation
 - The project must be uploaded on the portal web page before the written test
 - Use the "Materiale" page on the portal
 - Enroll for the written exam in that session
- Each project can add from **-2.0** to **+6.0** marks to the final written evaluation

Assessment and Grading Criteria

- Marks may differ for different students within the same group, depending on the effort they put into the project and on the final presentation
- Once the final mark for a project has been obtained, there is **no** time limit to its validity
- ❖ Final evaluation
 - The written test plus the group project may amount up to $30+6 = 36$ out of 30
 - Marks larger or equal to **32** will be automatically converted into 30 with honor

Assessment and Grading Criteria

On-Off



❖ **Addendum** for potentially “weak” students

- Students who are unable to pass the exam with the previous rules may follow a **simplified** examination path
- This path includes an **on/off** written test
 - The test concentrates more on theory issues and it is **less** demanding on problem-solving topics and C/C++ implementations
 - No design, mainly analysis
 - It includes closed (automatically corrected) and open (manually corrected) questions
 - It lasts from 90 to 150 minutes

Assessment and Grading Criteria

- Each test
 - Includes a first part (Prof. Cabodi) and a second part (Prof. Quer)
- The two parts must be taken during the **same** session
- Each part includes from 5 to 15 questions
- Each part delivers a **fail** or **pass** grade
- The candidate will pass the exam **if and only if** he/she has received a **pass** grade on both parts



To give a rough idea, following the rules of the standard written tests, each part will deliver about 15 points with a pass grade equal to about 7 points

Assessment and Grading Criteria

- Partial results in the on/off exam are lost
 - A pass result in a part (first or second) without a pass result in the other (second or first) is lost
 - The on/off exam has to be taken and passed entirely in the same session
- Students taking the on/off exam will automatically give-up (lose) their project and its evaluation (even if already obtained)

Assessment and Grading Criteria

- The simplified on/off tests can also be used by candidate who already passed either the first (Prof. Cabodi) or the second (Prof. Quer) part **but** not the other
 - When a student obtained a pass evaluation (at least 7 points) in one of the two written tests, but he/she is unable to pass the other, he/she can pass the other taking the on/off exam
 - Students with two passes (one in the standard exam and the other in the on/off exam) will pass the exam with 18/30 and **any** project (delivered or not) will be automatically lost

Results so far

❖ Results for SDP for 3 academic years

- 2019-2020, 2020-2021, 2021-2022
- Evaluation date: 26.02.2023



Student
Funnel

Total Number of ...	Total	[%]
... students enrolled	549	100
... exam taken	431	79
... students who take the exam (at least once)	334	61
... students who never take the exam	215	39
... passed (on total = /549)	311	57
... passed (on who takes the exam = /334)		93
Average mark	23.6	

About 40% on/off
version

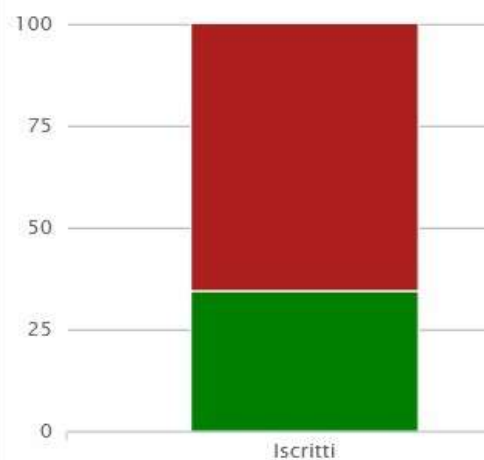
Results so far

01NYHOV - System and device programming

QUER STEFANO

A.A. 2020/21

Totale iscritti: 310 (134) - Superi: 107 (20 al primo anno)

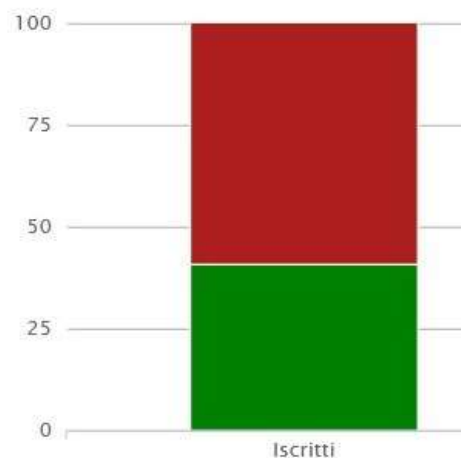


01NYHOV - System and device programming

QUER STEFANO

A.A. 2021/22

Totale iscritti: 318 (140) - Superi: 130 (41 al primo anno)

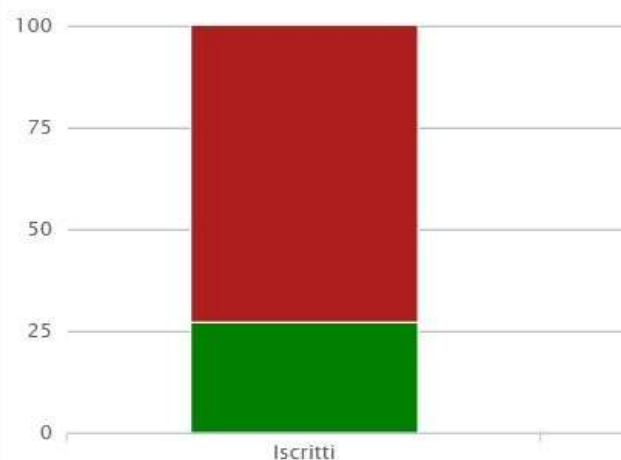


01NYHOV - System and device programming

QUER STEFANO

A.A. 2022/23

Totale iscritti: 320 (160) - Superi: 88 (35 al primo anno)



■ Esame non superato ■ Esame superato

To summarize

- ❖ During the course we will face both theory and practice problems

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 - Theory is when you know everything but nothing works

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 - Theory is when you know everything but nothing works
 - Practice is when everything works but no one knows why

To summarize

- ❖ During the course we will face both theory and practice problems
 - Theory is when you know everything but nothing works
 - Practice is when everything works but no one knows why
 - **In this class, theory and practice will be combined: Nothing will work and no one will know why**

(possibly) Albert Einstein, 1879-1955